Notes: Dix-Carneiro (Econometrica, 2014)
Trade Liberalization and Labor Market Dynamics

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1 Big picture

- **Question.** How does the labor market adjust dynamically after trade liberalization when workers face sector-switching costs, sector-specific experience, heterogeneous comparative advantage, and the possibility of leaving formal employment?
- **Setting.** The paper studies Brazil, using matched employer-employee data from RAIS. Brazil is a useful setting because it experienced major trade reform and has substantial formal-informal labor market movements.
- **Core object.** The paper estimates a structural dynamic equilibrium model of a small open economy with seven productive sectors and a residual non-formal/non-employment sector.
- **Central mechanism.** Trade liberalization changes sectoral output prices. These price changes move sectoral human-capital prices, which change workers' sector choices. Reallocation is slow because workers face direct mobility costs and because sector-specific experience is imperfectly transferable.
- **Main empirical result.** Median intersectoral mobility costs range from roughly 1.4 to 2.7 times annual average wages, depending on the destination sector. Costs are highly heterogeneous across workers.
- **Main simulation result.** A negative price shock to High-Tech Manufacturing generates large labor-market reallocation, but adjustment takes years. Depending on capital mobility, 95 percent of worker reallocation can take around 9 years or much longer.
- **Distributional result.** Workers initially employed in the adversely affected sector lose substantially, especially older and skilled workers.
- **Policy result.** A moving subsidy that compensates switching costs performs better than a retraining program in compensating losers, while adding only modest aggregate welfare adjustment costs.
- **Economic takeaway.** Long-run gains from trade can substantially overstate short- and medium-run welfare because workers and capital do not instantly move to their new efficient allocation.

2 Incremental contribution of the paper

- **From long-run trade gains to transition dynamics.** Standard trade models often compare initial and final equilibria. This paper explicitly studies the path in between and quantifies how slow adjustment reduces realized gains.
- **A structural bridge between trade and labor economics.** The model combines a small-open-economy trade environment with a dynamic Roy-style labor supply model featuring occupational/sector choice, human-capital accumulation, and worker heterogeneity.
- **Richer worker heterogeneity than earlier structural trade-adjustment models.** Unlike models with homogeneous workers and sector-specific preference shocks only, the paper allows comparative advantage across sectors by gender, education, age, sector-specific experience, and unobserved type.

- **Better counterfactual wages.** The paper emphasizes that observed sectoral average wages are poor counterfactual wage measures because of selection. Correcting for worker heterogeneity and selection sharply lowers estimated mobility costs relative to earlier approaches.
- **Formal residual-sector margin.** The model includes a Residual Sector that captures informal employment, self-employment, unemployment, or out-of-labor-force status, which is important in Brazilian data.
- **Capital mobility as a first-order transition force.** The simulations show that assumptions about physical capital mobility strongly shape transition speed and the magnitude of labor adjustment.
- **Distributional analysis.** The paper identifies which workers lose most from trade reform, not only which sectors contract in the long run.
- **Policy evaluation.** The model is used to compare retraining and moving-subsidy policies in the same equilibrium environment.

3 Data and measurement

3.1 Worker-level data: RAIS

- The core data come from RAIS, a matched employer-employee census of Brazil's formal labor market.
- Workers can be followed over time through individual identifiers, allowing the paper to observe sectoral transitions and wages while workers remain in formal employment.
- The estimation sample covers 1995-2005. Earlier RAIS observations, 1986-1994, are used to construct workers' initial sector-specific experience.
- Each worker-year is assigned to a single sector. If a worker holds multiple jobs, the job with the highest hourly wage is selected.
- Workers who leave formal employment enter a Residual Sector. This residual option is important because movements in and out of formal employment are frequent in Brazil.

Interpretation: The data are unusually well suited for dynamic sectoral adjustment because they contain worker panels, sector transitions, wages, and experience histories. The key limitation is that earnings outside formal employment are not observed.

3.2 Sector classification

The economy is divided into seven productive sectors plus a residual sector:

1. Agriculture and Mining;
2. Low-Tech Manufacturing;
3. High-Tech Manufacturing;
4. Construction;
5. Trade;
6. Transportation/Utilities/Communication;

7. Services;
8. Residual Sector.

Interpretation: The split between Low-Tech and High-Tech Manufacturing is central. Brazil has comparative advantage in relatively low-skill/low-tech activities, while High-Tech Manufacturing is treated as the import-competing sector affected by the simulated liberalization shock.

3.3 Aggregate data

- Sectoral value added comes from Brazilian National Accounts.
- Aggregate capital stock is taken from external capital-stock series.
- Because sector-level capital-stock data are unavailable, the estimation imposes efficient allocation of capital across sectors.
- The simulations relax this and compare perfect, no, and imperfect capital mobility.

4 Models and empirical specifications

4.1 Production side

Each productive sector has a representative competitive firm with a CES production function. Output in sector s depends on unskilled human capital, skilled human capital, and physical capital:

$$Y_t^s = p_t^s A_t^s \left[\alpha_t^{0,s}(H_t^{0,s})^{\epsilon_s} + \alpha_t^{1,s}(H_t^{1,s})^{\epsilon_s} + (1 - \alpha_t^{0,s} - \alpha_t^{1,s})(K_t^s)^{\epsilon_s}\right]^{1/\epsilon_s}. \tag{1}$$

- $H_t^{0,s}$ and $H_t^{1,s}$ are unskilled and skilled human capital in sector s.
- K_t^s is physical capital.
- $\alpha_t^{0,s}$ and $\alpha_t^{1,s}$ are time-varying factor weights.
- Competitive firms demand factors until marginal products equal factor prices.

Economic message: Sectoral price shocks affect the demand for skilled and unskilled human capital, which changes equilibrium human-capital prices and therefore workers' sector choices.

4.2 Worker dynamic problem

Workers choose the sector that maximizes the expected present value of utility:

$$V_a(\Omega_{iat}) = \max_{s\in\{0,1,\dots,7\}} V_a^s(\Omega_{iat}). \tag{2}$$

For a productive sector s, the value includes wages, nonpecuniary preferences, idiosyncratic shocks, and mobility costs:

$$V_a^s(\Omega_{iat}) = w^s(\Omega_{iat}) + \tau_s + \eta_{it}^s - Cost_{s_{t-1},s}(\Omega_{iat}) + \rho E[V_{a+1}(\Omega_{ia+1,t+1})]. \tag{3}$$

- Workers are forward looking.
- They accumulate sector-specific experience.
- They face costs when switching sectors.
- They can also choose a Residual Sector, interpreted as non-formal employment, unemployment, self-employment, or out of the formal labor force.

4.3 Human capital production

A worker's sector-specific human capital depends on observable characteristics, sector-specific experience, unobserved comparative advantage, and idiosyncratic shocks:

$$h^{e,s}(\Omega_{iat}) = \exp\left(X_i\beta^s + \sum_k \beta_k^s Exper_{ikt} + \theta_i^s + \varepsilon_{it}^s\right). \tag{4}$$

Interpretation: This specification is crucial because a worker's observed wage in one sector is not the wage she would necessarily earn in another sector. The model estimates counterfactual wages by correcting for observed and unobserved comparative advantage.

4.4 Mobility costs

The cost of moving from sector s to sector s' is modeled as:

$$Cost^{ss'}(\Omega_{iat}) = \exp\left(\varphi^{ss'} + \kappa_1 Female_i + \sum_j \kappa_j Educ_{ij} + \kappa_5(a-25) + \kappa_6(a-25)^2 + \lambda_i\right). \tag{5}$$

- Remaining in the same sector is costless.
- Moving into the Residual Sector is normalized to be costless.
- Costs vary by worker observables and unobserved type.

Economic message: Mobility costs summarize workers' inability to arbitrage sectoral wage differentials. They may include psychological costs, geographic mobility costs, search frictions, hiring/firing costs, or informational frictions.

5 Identification and empirical design

5.1 Indirect inference

The model is estimated by Indirect Inference. The idea is:

1. Estimate auxiliary models in the real data.
2. Simulate the structural model for a candidate parameter vector.
3. Estimate the same auxiliary models in the simulated data.
4. Choose structural parameters that make simulated auxiliary coefficients close to real-data auxiliary coefficients.

5.2 Auxiliary models

The auxiliary models include:

- sector-specific log-wage regressions;
- sector-specific residual wage variances;
- regressions for yearly log-wage changes;
- sectoral choice linear probability models;

- transition-rate models across sectors;
- return-to-origin-sector regressions after a switch;
- longer-run sectoral choice and years-spent-in-sector regressions;
- wage-bill and capital-income share targets.

Interpretation: The estimation is not identified by one regression. It is identified by the joint requirement that the model reproduce wages, wage dispersion, sectoral choices, transition rates, persistence, and factor income shares.

5.3 Identification intuition

- **Human capital parameters** are identified from wage levels and wage regressions, after solving the model to account for selection into sectors.
- **Mobility costs** are identified from transition rates conditional on wage differentials and worker characteristics.
- **Preference shocks** help explain two-way gross flows that are larger than net flows.
- **Residual-sector value** is identified from choices into and out of formal employment and from wage opportunities in productive sectors.
- **Unobserved types** are identified by persistent differences in wages and sectoral choices.

Core identification logic: If workers do not move despite large model-implied counterfactual wage gains, the model attributes this to mobility costs, persistent sector preferences, or non-transferable human capital. The auxiliary models help separate these channels.

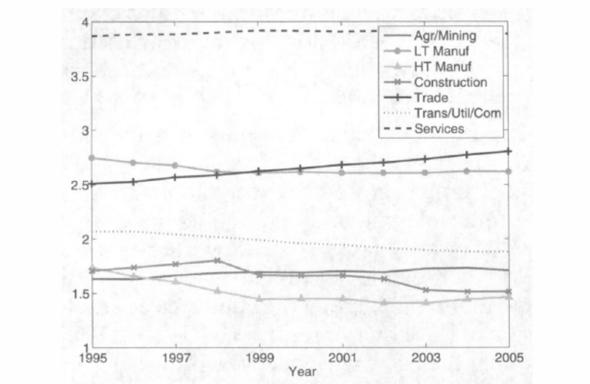
6 Tables and figures

6.1 Figures 1 and 2: Employment shares and wage differentials

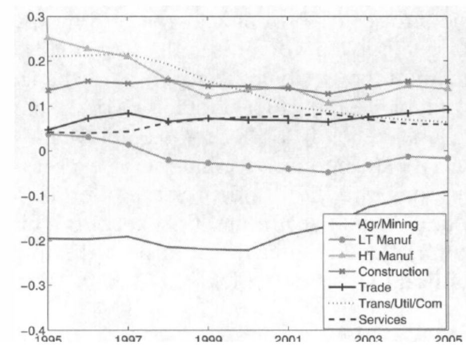
Read:

- Figure 1 shows the evolution of log employment shares across sectors in the formal labor market.
- Figure 2 shows large and persistent wage differentials across sectors.
- High-Tech Manufacturing and Transportation/Utilities/Communication are high-wage sectors; Agriculture/Mining is low wage.
- Net employment shares move slowly, even though gross flows are much larger.

Economic message: The data show both wage gaps and limited net reallocation. The model is designed to explain why workers do not simply move immediately toward high-wage sectors.



(a) Figure 1: Log-employment shares



(b) Figure 2: Wage differentials

6.2 Table I: Costs of mobility

Read:

- Median costs of moving from a formal sector into another formal sector range from about 1.4 to 2.7 annual average wages.
- Moving into High-Tech Manufacturing is especially costly.
- Moving from the Residual Sector into formal sectors is much more costly in unconditional terms.
- Conditional on actual switching, incurred mobility costs are much smaller and sometimes negative once nonpecuniary preferences and shocks are included.

Economic message: Large mobility costs are compatible with observed worker movement because the workers who move are selected: they receive favorable shocks, face lower costs, or have strong preferences for the destination sector.

TABLE I

COSTS OF MOBILITY

Median Cost of Entry Into ↓	From a Formal Sector	From the Residual Sector
A. Costs in Terms of Wages[a]		
Agr/Mining	1.43	8.93
LT Manuf	1.55	9.30
HT Manuf	2.70	10.62
Construction	1.40	9.24
Trade	1.38	8.96
Trans/Util	1.90	9.96
Services	2.17	9.17
B. Costs in Terms of Wages, Conditional on Switching[b]		
Agr/Mining	−0.66	1.39
LT Manuf	−0.13	1.74
HT Manuf	0.60	1.68
Construction	−0.44	1.44
Trade	−0.15	1.77
Trans/Util	0.23	1.35
[illegible]	[illegible]	[illegible]

Table I: Costs of mobility

6.3 Tables II and III: Model fit for wages and sector choices

Read:

- Table II compares average hourly wages in the data and model.
- Table III compares average sectoral choices in the data and model.
- The model reproduces the large Residual Sector and the dominance of Services among formal sectors.

Economic message: The model fits important unconditional moments before it is used for counterfactual liberalization experiments.

Sector	Data	Model
Agr/Mining	3.94	3.77
Low-Tech	5.77	5.68
High-Tech	11.66	12.08
Construction	4.99	4.97
Trade	4.53	4.56
Trans/Util	9.07	9.08
Services	9.29	9.47

(a) Table II: Wages

1995–2005: DATA VERSUS MODEL

	Data	Model
Residual	40.00	40.45
Agr/Mining	3.26	3.56
Low-Tech	8.38	8.54
High-Tech	2.70	2.60
Construction	3.12	3.15
Trade	8.64	8.62
Trans/Util	4.28	4.23
Services	29.61	28.85

(b) Table III: Sectoral choices

6.4 Table IV and Figure 3: Transition rates and goodness of fit

Read:

- Table IV compares transition rates across sectors in the data and simulated model.
- The diagonal entries are high, showing strong sectoral persistence.
- The Residual Sector has large flows into and out of formal sectors.
- Figure 3 plots standardized auxiliary-model coefficients in simulated data against actual data, with a close fit around the 45-degree line.

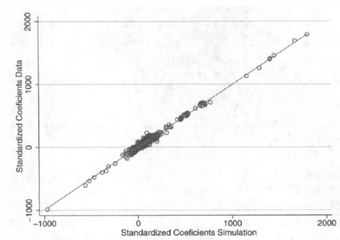
Economic message: The model captures both persistence and gross flows. This matters because mobility costs are disciplined by transition rates, not only by wage levels.

TABLE IV

TRANSITION RATES (%) 1995–2005: DATA VERSUS MODEL

	Residual	Agr/Mining	LT	HT	Const	Trade	Trans/Util	Services
A. Data								
Residual	79.80	1.64	2.73	0.58	2.02	3.64	1.05	8.54
Agr/Mining	17.39	75.93	2.26	0.55	0.77	0.92	0.63	1.56
LT	13.95	0.80	79.31	0.78	0.64	1.70	0.50	2.32
HT	10.49	0.60	2.25	81.26	0.63	1.37	0.52	2.88
Const	25.90	0.72	1.55	0.56	63.28	1.68	1.07	5.23
Trade	16.23	0.32	1.56	0.41	0.60	76.63	0.88	3.38
Trans/Util	11.02	0.38	0.73	0.23	0.69	1.30	83.08	2.57
Services	9.98	0.17	0.64	0.28	0.52	0.91	0.45	87.05
B. Model								
Residual	80.25	1.87	2.69	0.59	1.62	3.55	0.98	8.46
Agr/Mining	18.31	75.10	1.27	0.38	1.11	1.19	0.81	1.83
LT	14.90	0.52	79.86	0.64	0.63	1.27	0.64	1.54
HT	13.37	0.28	1.66	81.19	0.57	1.04	0.48	1.41
Const	19.08	1.14	2.62	1.03	68.95	1.95	1.47	3.75
Trade	16.23	0.47	1.61	0.43	0.94	77.05	0.88	2.38
Trans/Util	13.85	0.45	0.49	0.16	0.50	0.80	82.67	1.08
Services	10.33	0.23	0.57	0.14	0.45	0.76	0.37	87.15

(a) Table IV: Transition rates



	$\hat{\beta}_{\text{Data}}$
$\hat{\beta}_{\text{Sim}}$	0.9963
	(0.003)
Constant	0.000
	(0.000)
Observations	2,774
R^2	0.9814

FIGURE 3.—Goodness of Fit—Auxiliary models' standardized coefficients obtained in the *actual* data plotted versus those obtained in the *simulated* data. A perfect model fit would lead to all the points over the 45° line. The regression shown in the right hand side is weighted by the inverse of standard errors of the coefficients obtained with the *actual* data.

(b) Figure 3: Goodness of fit

6.5 Table V: Expenditure shares

Read:

- Table V reports Cobb-Douglas expenditure shares used to close the model in the simulations.
- Services has the largest expenditure share.
- The shares are used to determine consumption demand and price indexes.

Economic message: The expenditure shares matter because non-tradable sector prices adjust endogenously in the counterfactual economy.

TABLE V

EXPENDITURE SHARES

Agriculture/Mining	0.08
LT Manuf	0.07
HT Manuf	0.09
Construction	0.05
Trade	0.11
Trans/Util	0.12
Services	0.48

Table V: Expenditure shares

6.6 Figures 4 and 5: Dynamics under perfect and no capital mobility

Read:

- Figure 4 simulates a 30 percent adverse price shock to High-Tech Manufacturing when physical capital is perfectly mobile.
- Figure 5 repeats the shock when physical capital is sector-specific and cannot move.
- Under perfect capital mobility, High-Tech Manufacturing contracts dramatically because capital and labor both leave.
- Under no capital mobility, the sector survives because fixed capital raises the marginal product of remaining labor.

Economic message: Capital mobility fundamentally changes the transition path. Trade-induced labor adjustment cannot be understood by modeling labor frictions alone.

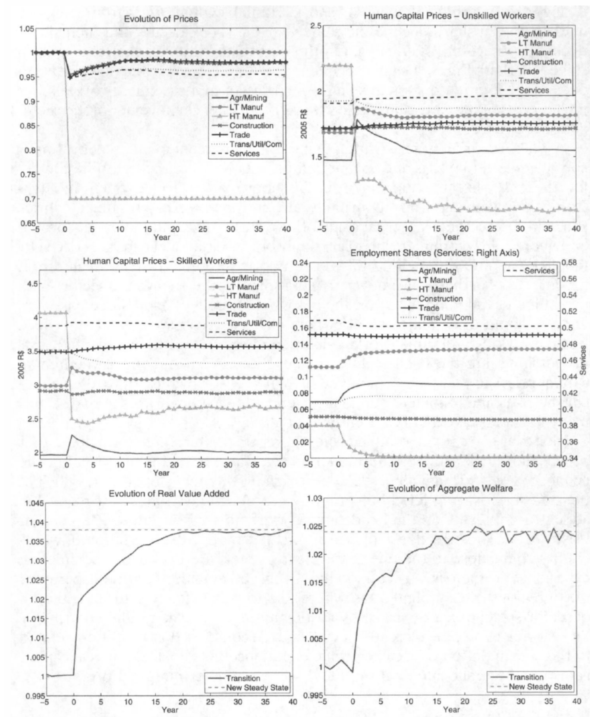


FIGURE 4.—Dynamics under *Perfect Capital Mobility* following the adverse price shock in the High-Tech Manufacturing sector illustrated in the left upper panel. The prices of the non-tradeable sectors adjust in equilibrium. The evolution of human capital prices, employment shares, real value added, and aggregate welfare following the shock are subsequently displayed in that order.

(a) Figure 4: Perfect capital mobility

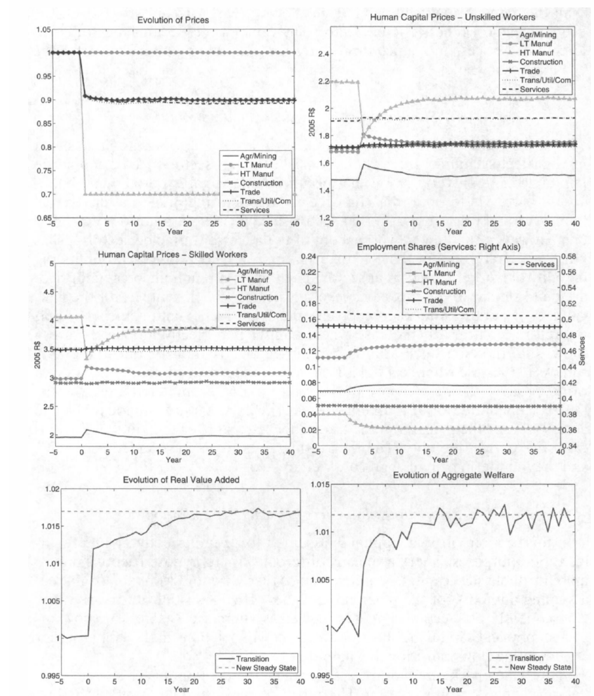


FIGURE 5.—Dynamics under *No Capital Mobility* following the adverse price shock in the High-Tech Manufacturing sector illustrated in the left upper panel. The prices of the non-tradeable sectors adjust in equilibrium. The evolution of human capital prices, employment shares, real value added, and aggregate welfare following the shock are subsequently displayed in that order.

(b) Figure 5: No capital mobility

6.7 Figure 6: Dynamics under imperfect capital mobility

Read:

- Figure 6 assumes imperfect capital mobility, with at most 10 percent of each sector's capital moving per year.
- Labor adjustment becomes much slower than in the perfect- or no-capital-mobility cases.
- The economy initially behaves like the no-capital-mobility case, then gradually moves toward the perfect-capital-mobility long-run allocation.

Economic message: Slow capital movement can amplify labor-market frictions and prolong the transition for decades.

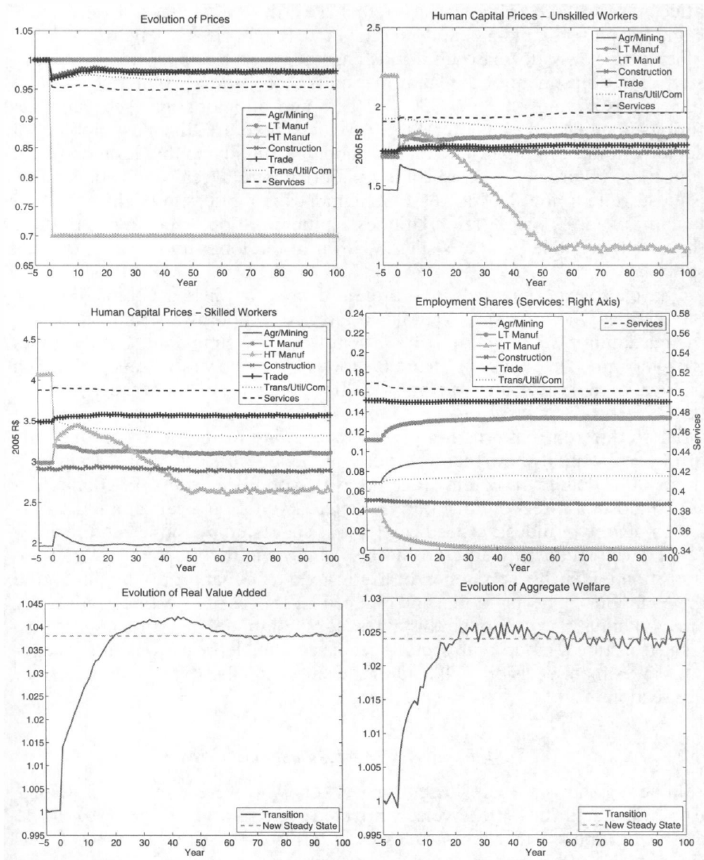


FIGURE 6.—Dynamics under *Imperfect Capital Mobility* (maximum rate of 10% per year) following the adverse price shock in the High-Tech Manufacturing sector illustrated in the left upper panel. The prices of the non-tradeable sectors adjust in equilibrium. The evolution of human capital prices, employment shares, real value added, and aggregate welfare following the shock are subsequently displayed in that order.

Figure 6: Imperfect capital mobility

6.8 Tables VI and VII: Welfare losses and adjustment costs

Read:

- Table VI reports welfare changes for workers initially employed in High-Tech Manufacturing.
- Losses are largest under perfect capital mobility and smallest under no capital mobility.
- Older and skilled High-Tech workers experience the largest losses.
- Table VII shows that adjustment costs can absorb 11 to 26 percent of the long-run gains from trade depending on capital mobility.

Economic message: Trade reform generates aggregate gains, but the timing and distribution of those gains are crucial. The path to the long run is costly.

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TABLE VI

WELFARE CHANGES (IN %) OF WORKERS WHO WERE EMPLOYED IN HT MANUFACTURING THE YEAR BEFORE THE SHOCK[a]

	Perfect Capital Mobility	No Capital Mobility	Imperfect Capital Mobility
Overall	−9.9	−4.9	−6.7
	By Demographics		
Old/Unskilled	−13.8	−6.4	−8.7
Old/Skilled	−22.5	−11.0	−15.3
Young/Unskilled	−3.8	−2.2	−2.6
Young/Skilled	−6.5	−3.5	−4.3

[a]Shock of 30% in the price of High-Tech Manufacturing. Imperfect mobility of capital: maximum rate of 10% per year. Old: 45 to 59 years old the year before the shock. Young: 25 to 44 years old the year before the shock.

(a) Table VI: Welfare changes

TABLE VII

ADJUSTMENT COSTS[a]

	Long Term Gain (%)	Adjustment Cost (%)
Perfect Capital Mobility	2.4	17.2
Imperfect Capital Mobility 1	2.4	18.2
Imperfect Capital Mobility 2	2.4	26.0
Imperfect Capital Mobility 3	2.4	15.9
No Capital Mobility	1.2	11.0

[a]Shock of 30% in the price of High-Tech Manufacturing. Imperfect Capital Mobility 1: maximum rate of 10% per year. Imperfect Capital Mobility 2: maximum rate of 5% per year. Imperfect Capital Mobility 3: maximum rate of 30% per year.

(b) Table VII: Adjustment costs

6.9 Tables VIII and IX: Labor-market policies

Read:

- Table VIII compares no policy, retraining policies, and moving subsidies for workers initially employed in High-Tech Manufacturing.
- Retraining policies only slightly reduce losses and can even hurt some young workers through equilibrium wage effects.
- Moving subsidies substantially reduce losses, especially when they cover the full switching cost.
- Table IX shows that moving subsidies raise aggregate adjustment costs only modestly relative to no policy.

Economic message: Policies that directly reduce switching frictions perform better than retraining policies in this model because the main estimated friction is the cost of moving, not the lack of transferable experience.

THE YEAR BEFORE THE SHOCK—DIFFERENT LABOR MARKET POLICIES[a]

	No Policy	Retraining 1	Retraining 2	Moving Subsidy 1	Moving Subsidy
Overall	−9.9	−9.7	−9.1	−1.4	−6.1
			By Demographics		
Old/Unskilled	−13.8	−12.2	−10.8	2.1	−6.6
Old/Skilled	−22.5	−22.2	−19.5	−8.7	−17.1
Young/Unskilled	−3.8	−4.2	−3.8	2.1	−0.9
Young/Skilled	−6.5	−6.9	−7.7	−2.3	−4.7

[a]Shock of 30% in the price of High-Tech Manufacturing. Perfect Physical Capital Mobility. Old: 45 to 59 years ol

(a) Table VIII: Welfare under policies

TABLE IX

WELFARE ADJUSTMENT COSTS (IN %) UNDER DIFFERENT LABOR MARKET POLICIES[a]

No Policy	17.2
Retraining Policy 1	17.8
Retraining Policy 2	20.8
Moving Subsidy 1	19.6
Moving Subsidy 2	17.6

[a]Shock of 30% in the price of High-Tech Manufacturing. Perfect Physical Capital Mobility.

(b) Table IX: Policy adjustment costs

6.10 Table X: Five-year elasticities

Read:

- Table X reports five-year wage and employment elasticities in High-Tech Manufacturing.
- The employment response is large, especially under perfect capital mobility.
- Under no or imperfect capital mobility, the model generates medium-run elasticities similar to estimates from the US literature.

Economic message: Mobility costs alone do not fully explain the lack of reallocation in developing-country evidence. Incomplete tariff pass-through, productivity responses, and capital frictions may also matter.

TABLE X

FIVE-YEAR PRICE ELASTICITIES OF WAGES AND EMPLOYMENT IN HIGH-TECH MANUFACTURING[a]

	Perfect Capital Mobility	No Capital Mobility	Imperfect Capital Mobility (10%)
Wage	1.3	0.3	0.6
Employment	2.9	1.3	1.9

[a]Shock of 30% in the price of High-Tech Manufacturing.

Table X: Five-year price elasticities

6.11 Appendix parameter tables

Read:

- Appendix Tables A.I-A.VII report estimated production, human-capital, residual-sector, mobility-cost, preference, type-specific, and type-probability parameters.
- These tables show the structural objects behind the counterfactual simulations.
- They are useful mainly for understanding how the model decomposes adjustment into production-side elasticities, human-capital transferability, mobility costs, and unobserved worker types.

TABLE A.I

PRODUCTION FUNCTION ESTIMATES[a]

	Agr/Mining	LT Manuf	HT Manuf	Construction	Trade	Trans/Util	Services
ϵ	0.1206	0.0681	0.4546	0.0828	0.2963	0.3909	0.1154
	(0.0247)	(0.0316)	(0.0259)	(0.0299)	(0.0461)	(0.0260)	(0.0253)
a_0	0.2529	0.4464	0.3380	0.3362	0.3251	0.4148	0.1844
(Intercept Unskilled)	(0.0153)	(0.0112)	(0.0085)	(0.0185)	(0.0145)	(0.0113)	(0.0081)
b_0	−0.00317	−0.0119	−0.0033	0.0000	0.0033	−0.0141	0.0029
(Slope Unskilled)	(0.0010)	(0.0018)	(0.0014)	(0.0019)	(0.0017)	(0.0018)	(0.0011)
a_1	0.1392	0.3062	0.6009	0.1651	0.4180	0.5168	0.5774
(Intercept Skilled)	(0.0165)	(0.0199)	(0.0113)	(0.0183)	(0.0277)	(0.0138)	(0.0141)
b_1	−0.00192	−0.0016	−0.0030	−0.0026	0.0004	−0.0020	0.0100
(Slope Skilled)	(0.0012)	(0.0021)	(0.0020)	(0.0019)	(0.0024)	(0.0022)	(0.0020)

[a]Standard errors in parentheses.

(a) Table A.I: Production function

TABLE A.II

HUMAN CAPITAL PRODUCTION FUNCTIONS: PARAMETER ESTIMATES[a]

	Agr/Mining	LT Manuf	HT Manuf	Construction	Trade	Trans/Util	Services
β_1^s: Female	−0.3485	−0.2878	−0.3325	−0.5155	−0.2798	−0.3507	−0.2497
	(0.0174)	(0.0103)	(0.0143)	(0.0193)	(0.0123)	(0.0143)	(0.0112)
β_2^s: $I(\text{Educ} = 2)$	0.1589	0.3104	0.3114	0.2444	0.3666	0.3090	0.3072
	(0.0127)	(0.0106)	(0.0118)	(0.0135)	(0.0119)	(0.0128)	(0.0088)
β_3^s: $I(\text{Educ} = 4)$	0.9600	0.7994	0.7998	0.5539	0.7217	0.7169	0.8484
	(0.0236)	(0.0134)	(0.0135)	(0.0201)	(0.0166)	(0.0149)	(0.0125)
β_4^s: (age − 25)	0.0370	0.0204	0.0203	0.0171	0.0106	0.0246	0.0188
	(0.0016)	(0.0015)	(0.0014)	(0.0016)	(0.0010)	(0.0014)	(0.0012)
β_5^s: $(\text{age} - 25)^2$	−0.0009	−0.0007	−0.0006	−0.0005	−0.0003	−0.0005	−0.0004
	(0.00005)	(0.00004)	(0.00005)	(0.00005)	(0.00002)	(0.00005)	(0.00004)
β_6^s: $\text{Exper}_{\text{Agr/Min}}$	0.1329	0.0227	0.0533	0.0114	0.00903	0.0679	0.0057
	(0.0022)	(0.0063)	(0.0050)	(0.0042)	(0.0081)	(0.0044)	(0.0067)
β_7^s: Exper_{LT}	0.0524	0.10242	0.0426	0.0260	0.0337	0.0371	−0.0083
	(0.0043)	(0.0024)	(0.0028)	(0.0038)	(0.0039)	(0.0038)	(0.0044)
β_8^s: Exper_{HT}	0.0619	0.0637	0.0978	0.0492	0.0562	0.0659	0.0207
	(0.0045)	(0.0028)	(0.0023)	(0.0040)	(0.0044)	(0.0037)	(0.0041)
β_9^s: $\text{Exper}_{\text{Const}}$	0.0407	0.025	0.0398	0.0988	0.0097	0.0429	−0.0007
	(0.0052)	(0.0057)	(0.0040)	(0.0028)	(0.0061)	(0.0043)	(0.0062)
β_{10}^s: $\text{Exper}_{\text{Trade}}$	0.0343	0.0478	0.041	0.0206	0.0913	0.0572	0.0057
	(0.0058)	(0.0035)	(0.0042)	(0.0046)	(0.0030)	(0.0046)	(0.0043)
β_{11}^s: $\text{Exper}_{\text{T/U}}$	0.0648	0.0451	0.0453	0.0436	0.0368	0.1011	0.0145
	(0.0045)	(0.0086)	(0.0052)	(0.0047)	(0.0062)	(0.0021)	(0.0046)
β_{12}^s: $\text{Exper}_{\text{Serv}}$	0.052	0.0581	0.0403	0.0454	0.055	0.0532	0.0929

(b) Table A.II: Human capital

TABLE A.III

VALUE OF THE RESIDUAL SECTOR: PARAMETER ESTIMATES[a]

γ_0: Intercept	0.8437
	(0.0207)
γ_1: Female	−0.2885
	(0.0141)
γ_2: $I(\text{Educ} = 2)$	0.3445
	(0.0140)
γ_3: $I(\text{Educ} = 3)$	0.6490
	(0.0207)
γ_4: $I(\text{Educ} = 4)$	1.6476
	(0.0206)
γ_5: (age − 25)	0.0474
	(0.0023)
γ_6: $(\text{age} - 25)^2$	−0.0014
	(0.00007)
σ_0: SD of Shock	18.5353
	[illegible]

(a) Table A.III: Residual sector

TABLE A.IV

COSTS OF MOBILITY: PARAMETER ESTIMATES[a]

	$\varphi^{\text{Res},s}$	φ^{s}_{Out}	$\varphi^{s'}_{\text{In}}$
Agr/Mining	3.2385		1.846
	(0.0097)		(0.023
LT Manuf	3.2805	−0.0784	1.917
	(0.0098)	(0.0229)	(0.025
HT Manuf	3.4098	0.2073	2.451
	(0.0092)	(0.0226)	(0.026
Construction	3.2767	−0.2219	1.813
	(0.0098)	(0.0186)	(0.032
Trade	3.2445	−0.0561	1.838
	(0.0104)	(0.0250)	(0.027
Trans/Util	3.3520	0.1713	2.105
	(0.0105)	(0.0311)	(0.032
Services	3.2693	−0.1753	2.084
	(0.0124)	(0.0223)	(0.028
κ_1: Female		0.1136	
		(0.0125)	
κ_2: $I(\text{Educ} = 2)$		0.0101	
		(0.0010)	
κ_3: $I(\text{Educ} = 3)$		0.0464	
		(0.0139)	
κ_4: $I(\text{Educ} = 4)$		0.2955	
		(0.0207)	
κ_5: (age − 25)		0.0263	
		(0.0015)	
κ_6: $(\text{age} - 25)^2$		−0.0006	
		(0.00005)	

[a]Standard errors in parentheses.

(b) Table A.IV: Mobility costs

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TABLE A.V

NON-PECUNIARY PREFERENCE PARAMETER ESTIMATES[a]

τ_2: LT	0.4758
	(0.0391)
τ_3: HT	0.4164
	(0.0337)
τ_4: Const	−0.0675
	(0.0277)
τ_5: Trade	0.1052
	(0.0261)
τ_6: T/U	0.2530
	(0.0312)
τ_7: Services	0.7147
	(0.0311)
ν: Scale Parameter of Shock	2.1489
	(0.0285)

(a) Table A.V: Nonpecuniary preferences

TABLE A.VI

TYPE-SPECIFIC PARAMETERS: PARAMETER ESTIMATES[a]

	Type II	Type III
θ^0_h: Residual	0.8583	−0.8208
	(0.0236)	(0.0320)
θ^1_h: Agr/Mining	0.4991	−0.6376
	(0.0137)	(0.0169)
θ^2_h: LT	0.8015	−0.5304
	(0.0112)	(0.0141)
θ^3_h: HT	0.5818	−0.9909
	(0.0101)	(0.0341)
θ^4_h: Construction	0.8738	−0.3027
	(0.0128)	(0.0171)
θ^5_h: Trade	0.8355	−0.4148
	(0.0167)	(0.0192)
θ^6_h: Trans/Util	0.5216	−0.9100
	(0.0115)	(0.0209)
θ^7_h: Services	0.7878	−0.6563
	(0.0118)	(0.0151)
λ_h: Mobility Cost	−0.1906	0.1875
	(0.0171)	(0.0179)

(b) Table A.VI: Type-specific parameters

TABLE A.VII

TYPE PROBABILITY FUNCTION: PARAMETER ESTIMATES[a]

	Type II	Type III
π_h^0: Intercept	0.2424	0.4988
	(0.0371)	(0.0328)
π_h^1: $\text{Exper}_{\text{Agr/Min}}$	0.3579	0.4161
	(0.0392)	(0.0359)
π_h^2: Exper_{LT}	0.0827	0.1247
	(0.0254)	(0.0243)
π_h^3: Exper_{HT}	0.3224	0.1629
	(0.0375)	(0.0357)
π_h^4: $\text{Exper}_{\text{Const}}$	0.1724	0.0869
	(0.0325)	(0.0333)
π_h^5: $\text{Exper}_{\text{Trade}}$	0.0954	0.2215
	(0.0298)	(0.0255)
π_h^6: $\text{Exper}_{\text{T/U}}$	0.0678	−0.0612
	(0.0264)	(0.0262)
π_h^7: $\text{Exper}_{\text{Services}}$	0.1945	0.2481
	(0.0249)	(0.0220)
π_h^8: Female	0.0439	0.2479
	(0.0264)	(0.0388)
π_h^9: $I(\text{Educ} = 2)$	0.0223	0.1046
	(0.0243)	(0.0302)
π_h^{10}: $I(\text{Educ} = 3)$	0.5081	0.6507
	(0.0597)	(0.0595)
π_h^{11}: $I(\text{Educ} = 4)$	0.3388	0.2040
	(0.0526)	(0.0365)

[a]Standard errors in parentheses.

Table A.VII: Type probabilities

7 Short summary

- This note summarizes Dix-Carneiro (2014), which estimates a structural dynamic equilibrium model of Brazil's labor market to study the transitional and distributional effects of trade liberalization using matched employer-employee data from RAIS.
- The model features heterogeneous workers, sector-specific experience, costly sector switching, unobserved comparative advantage, overlapping generations, and a Residual Sector outside formal employment.
- The paper estimates mobility costs by Indirect Inference, matching wages, sector choices, transition rates, persistence, and factor-income shares.
- Estimated median costs of switching across formal sectors are large, ranging from roughly 1.4 to 2.7 times annual average wages.
- Sector-specific experience is imperfectly transferable, but mobility costs appear more important than experience loss for explaining slow adjustment.
- Counterfactual trade liberalization is modeled as a permanent fall in the price of High-Tech Manufacturing.

- Labor-market adjustment is large but slow; depending on capital mobility, adjustment can take many years or decades.
- Workers initially employed in the adversely affected sector lose substantially, especially older and skilled workers.
- Adjustment costs absorb a meaningful fraction of potential gains from trade.
- Moving subsidies perform better than retraining programs in compensating displaced workers in the model.

8 Conclusion

8.1 Core conclusion

Dix-Carneiro shows that trade liberalization should not be evaluated only by comparing the initial and final allocations. The transition path matters. When workers have sector-specific experience and face large switching costs, the economy may take many years to realize the gains from trade, and some workers may suffer large welfare losses even if aggregate welfare rises in the long run.

8.2 Economic intuition

The intuition is that trade reform changes relative prices, but workers cannot instantly reallocate. A worker in a contracting sector compares lower current wages against the cost of moving, the value of sector-specific experience, the possibility of future wage recovery, nonpecuniary preferences, and the residual option. Physical capital also matters: if capital moves slowly, wages and employment adjust differently than in a model with either perfectly mobile or fixed capital.

8.3 Incremental contribution revisited

The paper contributes by embedding a rich dynamic labor model inside a trade-liberalization counterfactual. Its key advance is not simply estimating a mobility cost; it estimates a full equilibrium transition path with worker heterogeneity, sector-specific experience, nonemployment, and different capital mobility regimes. This makes the paper especially useful for understanding why the long-run gains from trade may coexist with short-run losses and political resistance.

8.4 Implications for trade policy

The paper implies that trade policy should be paired with labor-market policy. If adjustment is slow, the gains from liberalization are delayed and unevenly distributed. Direct mobility support may be more effective than generic retraining when estimated frictions are mainly mobility costs rather than missing sector-specific skills.

8.5 Limitations

The model abstracts from several important margins: endogenous schooling decisions, firm-level reallocation within sectors, explicit search and matching, geographic migration, and a fully structural model of capital investment. The simulations also rely on assumptions about tariff pass-through and the size of the High-Tech price shock.

8.6 Takeaway

The enduring takeaway is that trade liberalization is not only a comparison between two equilibria. It is a dynamic labor-market process. The workers who bear the transition costs, the speed at which resources move, and the policies used to smooth adjustment are central to the welfare consequences of trade reform.